Sections: B,C & D

Linear Algebra (MT-1004)

Homework#37

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Question 1

Diagonalization of Quadratic Forms

Let $Q(x_1, x_2) = 5x_1^2 - 4x_1x_2 + x_2^2$.

(a) Diagonalize the matrix using the eigenvalue decomposition method.

(b Rewrite $Q(x_1, x_2)$ in terms of the eigenvalues and transformed variables.

Question 2

Application in Machine Learning

A quadratic form appears in the cost function of ridge regression:

$$f(w) = w^T Aw + b^T w,$$

where A is the covariance matrix.

(a) What conditions on A ensure that $f(w)$ has a unique minimum?

(b) For $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, verify if Asatisfies these conditions.